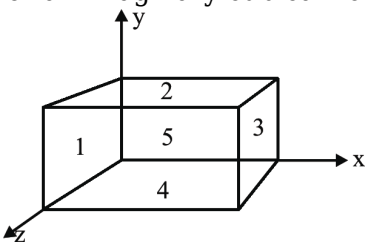
PART-1 : PHYSICS

SECTION-I (i)

1) In a region of space, the electric field $\vec{E} = E_0 x\hat{i} + E_0 y\hat{j}$. Consider an imaginary cubical volume of edge 'a' with its edges parallel to the axes of coordinates. Now,



(A) the total electric flux through the faces 1 and 3 is E_0a^3

(B) the charge inside the cubical volume is $2\varepsilon_0E_0a^3$

(C) the total electric flux through the faces 2 and 4 is $2E_0a^3$

(D) the charge inside the cubical volume is $\varepsilon_0E_0a^3$

2) At the equator on a spherical planet a body weighs one third of that at the pole. Radius of planet is R. The time period of revolution of the planet around its axis is equal to T = 60 min. (Take : $G = \frac{20}{3} \times 10^{-11}$ Nm^2/kg^2)

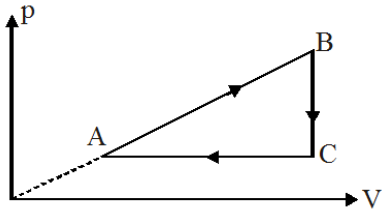
(A) The average density ρ of the planet is $\frac{\pi}{192} \times 10^6$ kg/m^3

(B) The height of geostationary satellite from the surface of planet is $\left(\sqrt[3]{\frac{3}{2}} - 1\right) R$

(C) At a latitude of 60°, the weight of the body is half the weight at pole

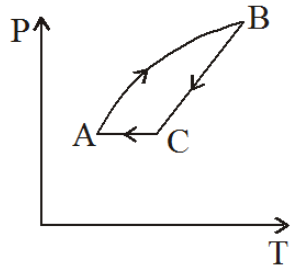
(D) The geostationary satellite can be in a plane passing through the poles.

3) A sample of ideal gas of constant mass is taken through the cyclic process shown in the figure. Plot this cyclic process on the (p,T), and (V,T) diagrams instead of the (p,V).

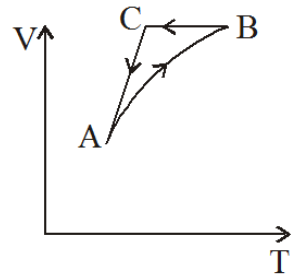


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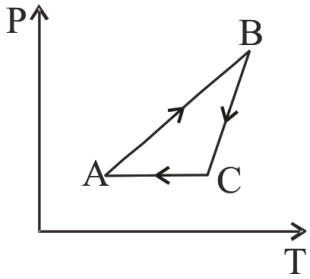
(A)



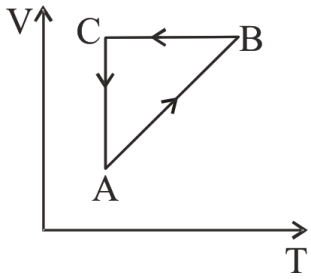
(B)



(C)



(D)



4) The potential difference between the target and the cathode of an X-ray tube is 50 kV. The current in the tube is 20 mA. Only 1% of the total energy is emitted as X-rays. (Take: hc = 1240 eV nm)

(A) The maximum frequency of the emitted X-rays is approx 1.2×10^{19} Hz.

(B) The heat should be removed at 900W to keep the target at constant temperature

(C) A wavelength of 0.2 nm can be emitted by this tube

(D) Electrons are being emitted at the rate of 1.25×10^{17}/sec by the cathode

5) Mark out the correct statement(s).

(A) Higher binding energy per nucleon means the nucleus is more stable.

(B) If the binding energy of nucleus were zero, then it would spontaneously break apart.

(C) Binding energy of a nucleus can be –ve.

(D) Binding energy of a nucleus is always +ve.

6) A particle of mass 6.6×10^{-30} kg starts (t = 0) moving on a straight line with velocity 10 m/s. Its velocity decreases with time, however rate of change of de-Broglie wavelength associated with

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particle remains constant at 10^{-4} m/s. (Take h = 6.6 × 10^{-34} J–s) :-

(A) Velocity of particle at t = 0.9s is 1 m/s

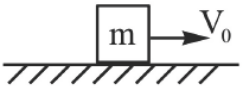
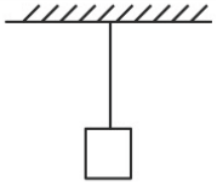
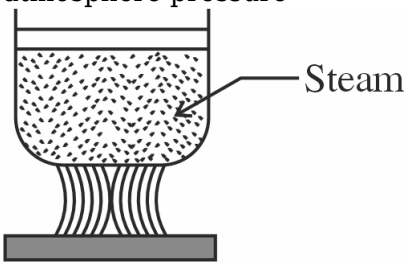
(B) Velocity of particle at t = 0.9s is 6 m/s

(C) Magnitude of retaration of particle at t = 0.9 s is 4 m/s^2

(D) Magnitude of retaration of particle at t = 0.9 s is 1 m/s^2

SECTION-I (ii)

1) In list-I, statement about heat given to the system, work and internal energy are given. Symbols have their usual meaning. Use first law of the thermodynamics in form ΔQ = ΔU + ΔW. Match the situations in list-II with statements in list-I.

List - I		**List - II**	
(I)	Block is the system. Assume no heat is lost to surrounding till the block comes to rest. 	(P)	ΔQ > 0
(II)	A block is released from rest from massless ideal non-conducting elastic string at natural length. After a few oscillations, block comes to rest. The string is the system. Assume the whole set-up is in vacuum. All joints are insulated. 	(Q)	ΔQ < 0
(III)	Water is slowly boiled in a closed container with a movable light piston. The container with its contents is the system. (Excluding piston)Don't ignore atmosphere pressure 	(R)	ΔW > 0
(IV)	A monatomic ideal gas undergoes a process PT^3 = constant. Gas is the system. Pressure increase during the process.	(S)	ΔU > 0
		(T)	ΔU < 0

(A) I → S;II → R;III → P;IV → R

(B) I → R;II → S;III → P;IV → S

(C) I → S;II → S;III → P;IV → Q,T

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(D) I → S;II → S;III → P;IV → S

2) Match list-I of nuclear process with list-II containing parent nucleus and one of end product of each process and then select the correct answer using codes given below the lists.

List-I		List-II	
(I)	Alpha decay	(P)	$_{91}Pa^{230} \rightarrow _{90}Th^{230} + ...$
(II)	$\beta^{\ominus}$ decay	(Q)	$_{30}Zn^{57} \rightarrow _{29}Cu^{56} + ...$
(III)	$\beta^{\oplus}$ decay	(R)	$_{1}H^{3} \rightarrow _{2}He^{3} + ...$
(IV)	neutron emission	(S)	$_{53}I^{137} \rightarrow _{53}I^{136} + ...$
		(T)	$_{94}Pu^{239} \rightarrow _{92}U^{235} + ...$

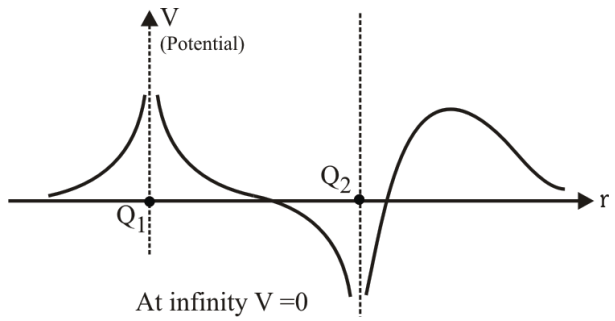
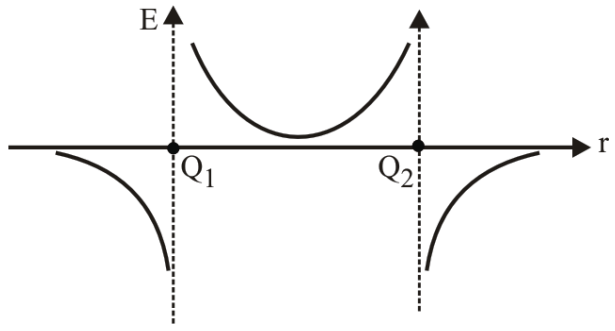
(A) I → T;II → S;III → P;IV → R

(B) I → T;II → P;III → Q;IV → S

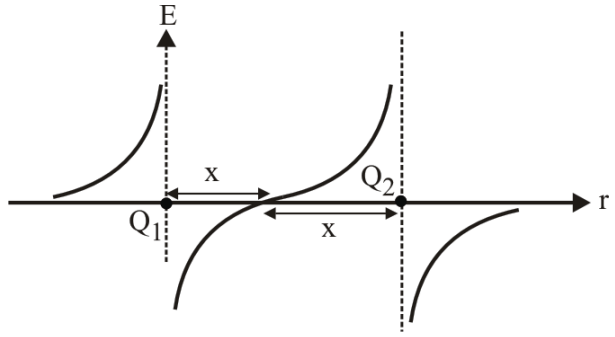
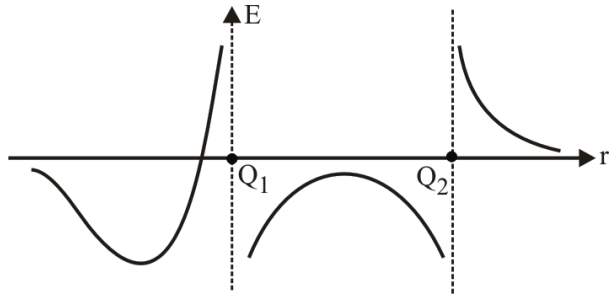
(C) I → T;II → R;III → P;IV → S

(D) I → Q;II → R;III → S;IV → P

3) As shown in List-I there are graphs of electric field (E) and potential (V) along the line joining charges Q_1 and Q_2 are drawn against distance (r) on x–axis for charges Q_1 and Q_2. Take potential at infinity equal to zero. [Take direction of E in rightward direction as positive]

	List-I		List-II
(I)		(P)	$Q_1 < 0, Q_2 > 0$
(II)		(Q)	$Q_1 > 0, Q_2 < 0$

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(III)		(R)	$\|Q_1\| > \|Q_2\|$
(IV)		(S)	$Q_1 < 0, Q_2 < 0$
		(T)	$\|Q_1\| = \|Q_2\|$

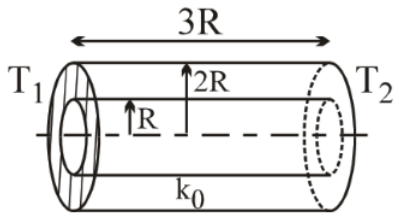
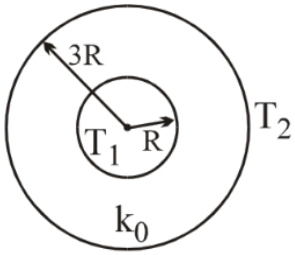
(A) I → Q;II → R;III → Q;IV → T

(B) I → Q;II → T;III → Q;IV → R

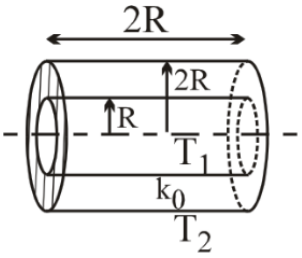
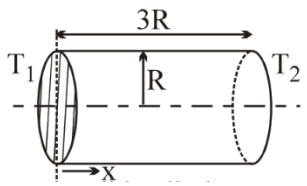
(C) I → Q;II → T;III → P;IV → Q

(D) I → S;II → T;III → Q;IV → T

4) Entries in list-I consists of diagrams of thermal conductors. The type of conductor & direction of heat flows are listed below. Entries in list-II consists of the magnitude of rate of heat flow belonging to any of the entries in list-I. If temperature difference in all the cases is $(T_1 - T_2)$, then match list

List-I		List-II	
(I)	 Thick cylindrical shell, flow along axis	(P)	$6\pi k_0 R(T_1 - T_2)$
(II)	 Thick spherical shell, radial flow	(Q)	$\frac{\pi k_0 R}{3\ell n 2}(T_1 - T_2)$

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(III)	 Thick cylindrical shell, radial flow	(R)	$\pi k_0 R(T_1 - T_2)$
(IV)	 Solid cylinder, flow along axis, variable k as $k = k_0(1 + x/(3R))$	(S)	$\frac{4\pi k_0 R}{\ell n 2}(T_1 - T_2)$
		(T)	None of these

(A) I → R;II → P;III → S;IV → Q

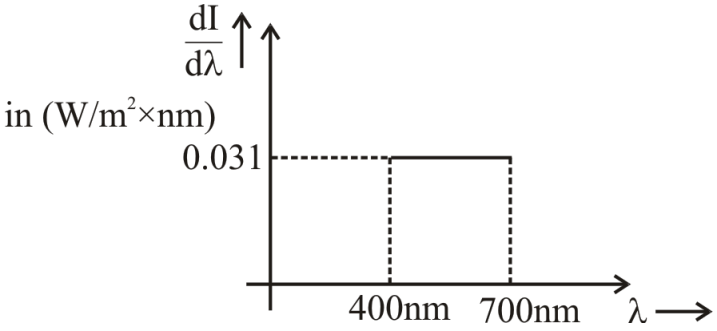
(B) I → P;II → R;III → R;IV → S

(C) I → Q;II → R;III → P;IV → R

(D) I → P;II → P;III → R;IV → Q

SECTION-II

1) Scientists have made a light source whose spectral emissive power $\frac{dI}{d\lambda}$ is constant over visible range. Here I is intensity and λ is wavelength. In other words, $\frac{dI}{d\lambda}$ graph is shown below. This beam of area $1m^2$ is incident on emitter plate of photoelectric tube. The collector plate is sufficiently positive so that tube is in saturation mode. Assume that each capable photon liberates 1 electron. If work function is 2eV, what is the current (in A) the tube? Round off to nearest integer.



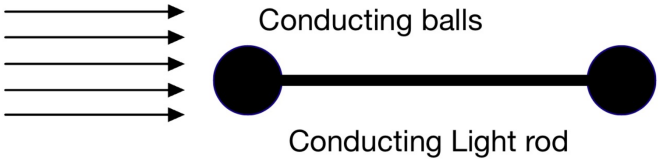
2) A hydrogen like atom is in an excited state '2n' can emit a maximum energy photon of 204eV. If it makes a transition to a quantum state 'n', a photon of energy 40.8eV is emitted. Calculate 'n'. Take ground state energy of hydrogen atom to be –13.6eV.

3) Starting from t = 0, activity of a radioactive sample is A_1 at time t_1 and A_2 at time t_2. If τ is average life of sample then the number of nuclei decayed in time $(t_2 - t_1)$ is $\beta(A_1 - A_2)$ in proper unit. Here β is an integer. Find β. (Given that t_1 = 5 sec, t_2 = 8 sec and τ = 7 sec)

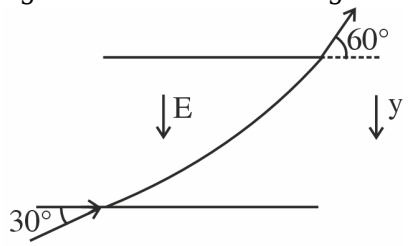
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4) The efficiency of a carnot engine is 0.6. It rejects total 20 J of heat. The work done by the engine (in kJ) is :

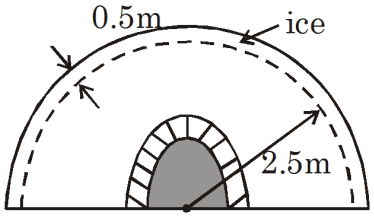
5) Two identical conducting balls of radius R are joined with light & conducting rod of length ℓ. This system placed in horizontal field E as shown. Now system tilted slightly and released. The time period of small oscillation is $P \times \pi\sqrt{\frac{m}{4\pi\epsilon_0 E^2 R}}$ then find P. (m = mass of each solid ball and >>R)



6) Find the magnitude of uniform electric field E in N/C (direction shown in figure) if an electron entering with velocity 100 m/s making 30° comes out making 60° (see figure), after a time numerically equal to $\frac{m}{e}$ of electron.



7) An igloo, a hemispherical enclosure built of ice (k = 1.67 J/m-s°C), has an inner radius of 2.50 m. The thickness of the ice is 0.5 m. At what rate must thermal energy be generated (in kW) to maintain the air inside the igloo at 5°C when the outside temperature is –40°C ? Ignore all thermal energy losses through the ground or by air currents. (Take : π = 3.14)



8) At the disposal of a physicist there are two thermal reservoirs, first is very hot with temperatures 227 °C and the second just hot with temperature of 77 °C. Physicist is told to heat the very hot body by extracting heat of 1000 J from just hot body. What is the minimum mechanical work (in Joule) he'll have to do? The heat capacities of both the bodies can be considered very large.

PART-2 : CHEMISTRY

SECTION-I (i)

1) In cubic lattice atom A occupy all cornors, atom B occupy alternate face centres and atom C occupy alternate edge centres. If a defect is created by removing all atoms present on one of line

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passing through opposite edge centres then which of the following formula is possible for new unit cell?

(A) ABC_2

(B) A_2B_2C

(C) A_2BC

(D) A_2BC_2

2) Select the correct statement(s) :

(A) Number of nodal planes and radial nodes in 5d orbital are 2 and 1 respectively.

(B) All four quantum number are obtained from schrodinger wave equation when applied to an atom.

(C) When an electron jumps from 4th excited state to ground state in a H-atom, maximum number of photons that may be emitted are 4.

(D) If the shortest wavelength of H-atom in Lyman series is x, then the longest wavelength in Balmer series of He^+ ion is $\frac{9x}{5}$.

3) Which of the following is/are correct on the basis of MOT ?

(A) Bond strength of CO > CO^+

(B) Bond length order : $O_2 > O_2^+ > O_2^-$

(C) Number of unpaired electrons order : O_2 > NO > CO

(D) Number of ABMO electrons order : $O_2 > N_2^- > Be_2$

4) Which of the following reaction(s) is/are written correctly (w.r.t. products)?

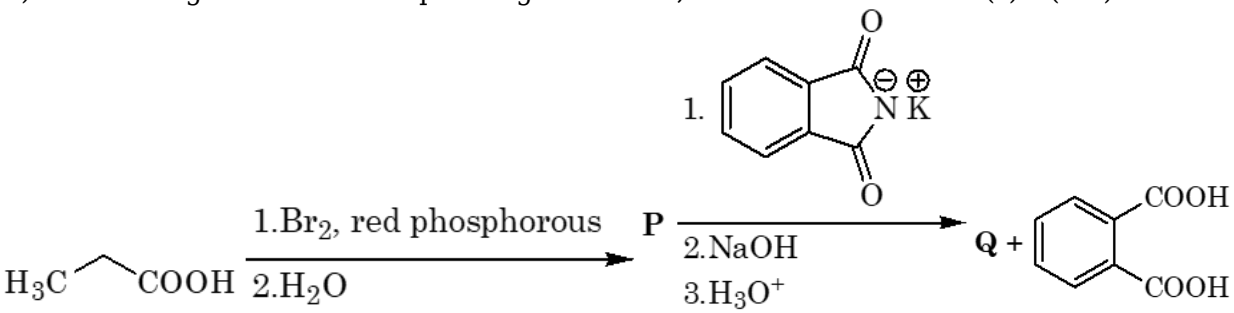
(A) $2FeSO_4 + H_2SO_4 + Cl_2 \rightarrow Fe_2(SO_4)_3 + 2HCl$

(B) $Na_2SO_3 + Cl_2 + H_2O \rightarrow Na_2SO_4 + 2HCl$

(C) $SO_2 + 2H_2O + Cl_2 \rightarrow H_2SO_4 + 2HCl$

(D) $I_2 + 6H_2O + \underset{\text{(excess)}}{5Cl_2} \rightarrow 2HIO_3 + 10HCl$

5) Considering the reaction sequence given below, the correct statement(s) is(are)



(A) P can be reduced to primary alcohol using $NaBH_4$.

(B) Treating P with conc.NH_4OH solution followed by acidification gives Q.

(C) Treating Q with a solution of $NaNO_2$ in aq.HCl liberates N_2.

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(D) P is more acidic than CH_3CH_2COOH.

6) Reaction in which aldehyde is/are obtained as product :

(A) $HC \equiv CH \xrightarrow[dil\ H_2SO_4]{HgSO_4}$

(B) $CH_3 - C \equiv CH \xrightarrow[(ii)H_2O_2/OH^{\ominus}]{(i)BH_3/THF}$

(C) $CH_3 - \underset{\underset{O}{\|}}{C} - Cl \xrightarrow[BaSO_4]{H_2/Pd}$

(D) $CH_3 - CH_2 - OH \xrightarrow[300^oC]{Cu}$

SECTION-I (ii)

1)

List-I		List-II	
(P)	$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$, $\Delta H > 0$	(1)	High pressure and low temperature will favour formation of products.
(Q)	$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$. $\Delta H > 0$	(2)	Increase in the amount of any of the product will definitely favour backward reaction.
(R)	$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $\Delta H > 0$	(3)	Increase in temperature as well as doubling the volume of reaction vessel must cause the equilibrium to shift in forward direction.
(S)	$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H < 0$	(4)	A decrease in temperature will cause decrease in the value of equilibrium constant.
		(5)	Increase in volume decrease the concentration of (each) products

Codes :

(A) P → 2,3,4;Q → 3,4;R → 2,3,4,5;S → 1,2,5

(B) P → 2,3,4,5;Q → 3,4;R → 2,3,4;S → 1,2,5

(C) P → 2,3,4,5;Q → 3,4;R → 2,3,4,5;S → 1,5

(D) P → 2,3,4;Q → 2,3,4;R → 2,3,4,5;S → 1,4,5

2) For the given galvanic cell at 25°C, Pt, H_2(g, 1 atm) | H^+(aq. 0.1M) || Cl^-(aq. 0.01 M) | Cl_2(g, 1 atm), match the list-I (change in concentration or quantity) with list-II (change in EMF).

(Given : $\frac{2.303\,R \times 298}{F}$ = 0.06V)

List-I	List-II

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(P)	pH of solution in anodic compartment change to pH = 5	(1)	– 0.03 V
(Q)	Concentration of ions in cathodic compartment increased by factor 10	(2)	0.03 V
(R)	Pressure of gas flown in anodic compartment increased by factor 10	(3)	– 0.06 V
(S)	Pressure of gas flown in cathodic compartment decreased by factor 10	(4)	0.24 V
		(5)	–0.24 V

(A) P → 5;Q → 4;R → 1;S → 2

(B) P → 4;Q → 3;R → 2;S → 1

(C) P → 3;Q → 4;R → 2;S → 1

(D) P → 5;Q → 3;R → 1;S → 2

3) Match the reactions (in the given stoichiometry of the reactants) in List-I with one of their products given in List-II and choose the correct option.

List-I		**List-II**	
(P)	$P_2O_3 + 3H_2O \rightarrow$	(1)	$P(O)(OCH_3)Cl_2$
(Q)	$P_4 + 3NaOH + 3H_2O \rightarrow$	(2)	H_3PO_3
(R)	$PCl_5 + CH_3COOH \rightarrow$	(3)	PH_3
(S)	$H_3PO_2 + 2H_2O + 4AgNO_3 \rightarrow$	(4)	$POCl_3$
		(5)	H_3PO_4

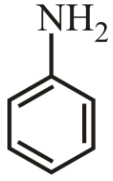
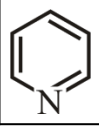
(A) P → 2;Q → 3;R → 1;S → 5

(B) P → 3;Q → 5;R → 4;S → 2

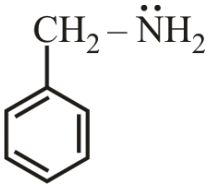
(C) P → 5;Q → 2;R → 1;S → 3

(D) P → 2;Q → 3;R → 4;S → 5

4)

List-I		**List-II**	
(P)		(1)	Conjugated acid is aromatic
(Q)		(2)	Most basic compared

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(R)		(3)	Lone pair of nitrogen involve in resonance
(S)		(4)	Conjugated acid is non aromatic
		(5)	Anti aromatic

(A) P → 1;Q → 3;R → 1;S → 3

(B) P → 2;Q → 3;R → 3;S → 1

(C) P → 2;Q → 4;R → 1;S → 2

(D) P → 3;Q → 4;R → 1;S → 2

SECTION-II

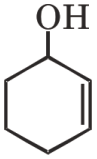
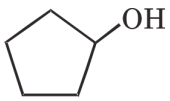
1) A solution of 0.2 mole KI (α = 100%) in 1000 gram water freezes at –0.2°C. If in this solution, 0.1 mole solid HgI_2 is added. Now the resulting solution freezes at ___kelvin.
(Given : water freezes at 273 K)

2) When 1 mole of A react with $\frac{1}{2}$ mole of B_2, $\left[A(g) + \frac{1}{2}B_2(g) \rightarrow AB(g)\right]$, 100 kcal heat is liberated. When 1 mole of A reacted with 2 mole of B_2, [$A(g) + 2B_2(g) \rightarrow AB_4(g)$], 200 kcal heat is liberated. When 1 mole of A completely reacted with excess of B_2 to form AB(g) and AB_4(g), 140 kcal heat is liberated. Calculate the moles of B_2(g) used.

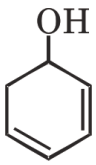
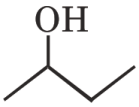
3) How many statement(s) is/are correct **(from the given below statements)**.
(i) In the presence of 3° Amine Homolytic cleavage of B_2H_6 take place.
(ii) Unpaired e^- of $\bullet CH_3$ & $\bullet CF_3$ is present in hybrid sp^3 orbital
(iii) On increasing bond strength , bond length always decreases.
(iv) All species in which 3c-4e^- type of Bridge bonds are available are non-planar.
(v) Due to formation of back bond angles are always increase.
(vi) On increasing size of anion Polarizability increase.

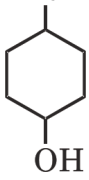
4) How many species will show redox reaction on hydrolysis?
XeF_2, XeF_4, XeF_6, XeO_3, XeO_2F_2, $XeOF_4$

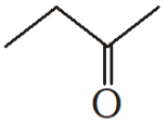
5) Find out number of alcohols, which can give immediate white turbidity with $ZnCl_2$ + Anhy.HCl

(1) Me_3C–OH (2)  (3)  (4) 

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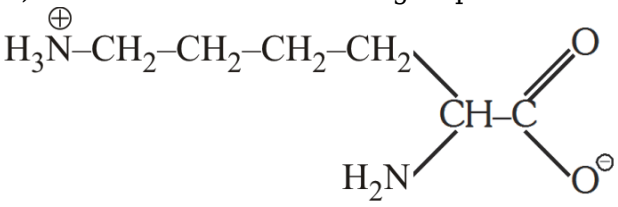
(5) $CH_3O–CH_2–OH$ (6) Ph – CH_2 – OH (7)  (8) 

(9) 

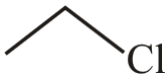
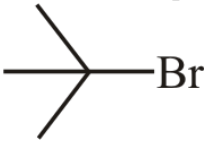
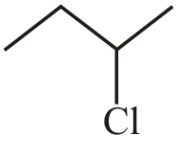
6)  + CH_3CHO $\xrightarrow[5^oC]{\overset{\ominus}{OH}}$

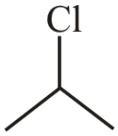
How many alcohol products will be obtained in above reaction? (excluding stereoisomerism)

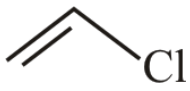
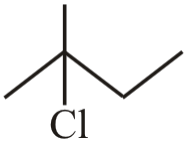
7) The total number of basic groups in the following form of lysine is :

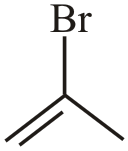
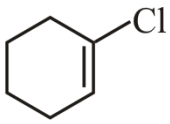


8) Number of R–X use in Gabril pthalimide reaction for preparation of amine?

(i)  (ii)  (iii) 

(iv)  (v)  (vi) Ph – CH_2 – Cl

(vii)  (viii) 

(ix)  (x) 

PART-3 : MATHEMATICS

SECTION-I (i)

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1) If the line $2x + \sqrt{6}y = 2$ touches the hyperbola $x^2 - 2y^2 = 4$, then point of contact is

(A) $\left(-2, \sqrt{6}\right)$

(B) $\left(-5, 2\sqrt{6}\right)$

(C) $\left(\frac{1}{2}, \frac{1}{\sqrt{6}}\right)$

(D) $\left(4, -\sqrt{6}\right)$

2) From point $P(\sqrt{6}, 2)$ a line is drawn meeting the circle $x^2 + y^2 = 1$ at points A & B. Then (PA + PB) cannot be

(A) 2

(B) 3

(C) 4

(D) 5

3) Which of the following statement is/are true

(A) Product of two irrational numbers is always irrational

(B) Sum of two irrational number is always irrational

(C) If a, b, c ∈ R then $a^2 + b^2 + c^2 - ab - bc - ca \geq 0$

(D) If a + b + c = 0 then $(ab + bc + ca)^2 = a^2b^2 + b^2c^2 + c^2a^2$

4) If A is a 3 × 3 matrix, then correct option/s is/are (where k is a non zero scalar)

(A) $|adj(kA)| = k^6|A|^2$

(B) $|adj(adjkA)| = k^{12}|A|$

(C) $\left|\frac{A}{|A|}\right| = 1$

(D) $|(kA)^{-1}| = \frac{1}{k|A|}$

5) Consider an ellipse E : $\frac{x^2}{16} + \frac{y^2}{4} = 1$

(A) Normal is drawn at a point P on E to intersect x and y axis at A and B respectively then locus of circumcentre of ΔOAB is $4x^2 + y^2 = 9$ (where O is the origin)

(B) Normal is drawn at a point P on E to intersect x and y axis at A and B respectively then locus of circumcentre of ΔOAB is $x^2 + 9y^2 = 4$ (where O is the origin)

(C) Area of quadrilateral formed by foci and end points of minor axes of the given ellipse is $8\sqrt{3}$ sq. units

(D) If a tangent is drawn at P on E to intersect the x-axis and y-axis at L and M respectively then minimum value of LM is 6 units

6) The value of x in [0, 2π] for which $2^{(2\sin x - 1)(\sin x - 1)} + 2^{(2\sin x + 1)(2 - \sin x)} = 9$, is

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(A) $\frac{5\pi}{6}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{2}$

(D) $\frac{7\pi}{6}$

SECTION-I (ii)

1) Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(I)	The system of equation $2x - y + z = 0$; $x - 2y + z = 0$ and $\lambda x - y + 2z = 0$ has non-trivial solutions then λ is greater than or equals to	(P)	0
(II)	The value of c for which the lines $x + y = 1$, $(c + 2)x + (c + 4)y = 6$ and $(c + 2)^2x + (c + 4)^2y = 36$ are concurrent is	(Q)	2
(III)	The equation $2x^2 - 3xy - py^2 + x + qy - 1 = 0$ represents two mutually perpendicular lines then $\|2(p + q)\|$ is	(R)	3
(IV)	The line $\lambda x + \mu y = 1$ is a normal to the circle $2x^2 + 2y^2 - 5x + 6y - 1 = 0$ then $\frac{(5\lambda - 6\mu)}{2}$ is	(S)	5
		(T)	6

(A) I → P,Q,R,S;II → Q;III → S;IV → Q

(B) I → Q;II → R,S;III → T;IV → P

(C) I → S,T;II → Q;III → S;IV → Q

(D) I → Q,S;II → R;III → T;IV → P

2) Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where a,b > 0 be a hyperbola in the xy-plane whose centre is C, one of the foci is F_1. Foot of perpendicular from F_1 upon line L : bx – ay = 0 is P. Triangle CF_1P is isosceles triangle whose area is 8 sq. units.

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(I)	The length of transverse axis of H is	(P)	$4\sqrt{2}$
(II)	The eccentricity of H is	(Q)	1
(III)	The slope of L is	(R)	4
(IV)	The distance between directrices of H is	(S)	$\sqrt{2}$
		(T)	8

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(A) I → S;II → T;III → Q;IV → P

(B) I → T;II → S;III → Q;IV → P

(C) I → T;II → S;III → Q;IV → R

(D) I → T;II → R;III → Q;IV → P

3) Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(I)	If the distance between the pair of parallel lines $x^2 + 2xy + y^2 - 8\lambda x - 8\lambda y - 9\lambda^2 = 0$ is $25\sqrt{2}$ then $\lvert\lambda\rvert$ equals	(P)	1
(II)	If the lines represented by the equation $\lambda(\lambda - 3)x^2 + 16xy + (\lambda + 1)y^2 = 0$ are perpendicular to each other, then λ equals	(Q)	2
(III)	The image of point A(1, 2) by the line mirror y = x is the point B and the image of B by the line mirror y = 0 is C and the image of C by the line mirror x + y = 0 is the point D(α, β) then ($\lvert\alpha\rvert + \lvert\beta\rvert$) equals	(R)	5
(IV)	Given ΔABC with unequal sides. P is the set of all the points which are equidistant from B and C and Q is the set of all the points which are equidistant from AB and AC, then n(P∩Q) equals	(S)	3
		(T)	7

(A) I → R;II → P;III → S;IV → Q

(B) I → P;II → R;III → S;IV → Q

(C) I → Q;II → P;III → S;IV → R

(D) I → R;II → P;III → Q;IV → S

4) For parabola $y^2 - 4y - x + 3 = 0$.

	List-I		**List-II**
(I)	Focus of parabola is	(P)	(0,0)
(II)	Extrimity of latus rectum is	(Q)	(–1,2)
(III)	Foot of directrix is	(R)	$\left(-\frac{3}{4}, 2\right)$
(IV)	Vertex of parabola is	(S)	$\left(-\frac{5}{4}, 2\right)$
		(T)	$\left(-\frac{3}{4}, \frac{5}{2}\right)$

(A) I → T;II → S;III → Q;IV → S

(B) I → R;II → T;III → Q;IV → S

(C) I → T;II → Q;III → P;IV → S

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(D) I → R;II → T;III → S;IV → Q

SECTION-II

1) Length of common chord of two circles of radii 3 and 4 which intersect orthogonally is □, then $\frac{24}{\ell}$ is

2) If adjA = $\begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$ and |A| > 0, then value of $\frac{1}{2}$ tr(A) is equal to
(where adj(A), tr(A) and |A| denotes adjoint, trace of matrix A and determinant of A respectively)

3) If a five digit number 2785x is divisible by 11, then the value of x is

4) Let □ = $(\log_3 4 + \log_2 9)^2 - (\log_3 4 - \log_2 9)^2$, $m = 0.8\left(1 + 9^{\log_3 8}\right)^{\log_{65} 5}$ and n is the value of x satisfy the equation $2^{(\log_2 3)^x} = 3^{(\log_3 2)^x}$ the determine $\frac{n\ell}{m}$

5) If $\begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & 18 & 2007 \\ 0 & 1 & 36 \\ 0 & 0 & 1 \end{bmatrix}$, then value of $\frac{(n+a)}{100}$ is

6) If $\begin{vmatrix} 1 & (\cos\alpha - \sin\alpha) & (\cos\alpha + \sin\alpha) \\ 1 & (\cos\beta - \sin\beta) & (\cos\beta + \sin\beta) \\ 1 & (\cos\gamma - \sin\gamma) & (\cos\gamma + \sin\gamma) \end{vmatrix} = k \begin{vmatrix} 1 & \cos\alpha & \sin\alpha \\ 1 & \cos\beta & \sin\beta \\ 1 & \cos\gamma & \sin\gamma \end{vmatrix}$, then k is

7) If the value of m for which the line $y = mx + 2\sqrt{5}$ touches the curve $16x^2 - 9y^2 = 144$ are roots of the equation
$x^2 - (a + b)x + k = 0$, then the value of |a + b + k| is equal to

8) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is such that it has least area but contains the circle $(x - 1)^2 + y^2 = 1$. If eccentricity of ellipse is e, then value of $3e^2$ is equal to

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ANSWER KEYS

PART-1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4	5	6
A.	A,B	A,B	A,B	A,C,D	A,B,D	A,D

SECTION-I (ii)

Q.	7	8	9	10
A.	C	C	D	A

SECTION-II

Q.	11	12	13	14	15	16	17	18
A.	2.80	2.00	7.00	0.03	2.00	100.00	7.06	428.57

PART-2 : CHEMISTRY

SECTION-I (i)

Q.	19	20	21	22	23	24
A.	B,C,D	C,D	C,D	A,B,C,D	B,C,D	A,B,C,D

SECTION-I (ii)

Q.	25	26	27	28
A.	A	B	D	A

SECTION-II

Q.	29	30	31	32	33	34	35	36
A.	272.85	1.10	2.00	2.00	6.00	6.00	2.00	4.00

PART-3 : MATHEMATICS

SECTION-I (i)

Q.	37	38	39	40	41	42
A.	D	A,B,C,D	C,D	A	A,C,D	A,B,C,D

SECTION-I (ii)

Q.	43	44	45	46
A.	A	B	A	D

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SECTION-II

Q.	47	48	49	50	51	52	53	54
A.	**5.00**	**2.50**	**2.00**	**2.00**	**2.00**	**2.00**	**4.00**	**2.00**

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SOLUTIONS

PART-1 : PHYSICS

1) $\phi = \vec{E}.\vec{A}$

ϕ_{in} = –ve and ϕ_{out} = +ve

$Q_{net} = \frac{q_{in}}{\varepsilon_0}$

$\phi_1 = 0, \phi_2 = E_0a^3, \phi_3 = E_0a^3, \quad \phi_4 = 0, \phi_5 = 0, \phi_6 = 0$

$Q_{net} = \phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6 = 2E_0a^3 \Rightarrow 2E_0a^3 = \frac{q_{in}}{\epsilon_0} \Rightarrow q_{in} = 2\varepsilon_0 E_0 a^3$

2) $g - \omega^2 R = \frac{g}{3}$

$\omega^2 R = \frac{2g}{3}$

$\left(\frac{2\pi}{60 \times 60}\right)^2 \times R = \frac{2}{3} \times \frac{GM}{R^2}$

$\frac{4\pi}{3600} \times \frac{3 \times 2}{2G} \frac{GM}{4/3R^3} = P$

$P = \frac{2\pi}{3600 \times 3600 \times 2 \times (20/3) \times 10^{-6}}$

$= \frac{10^6 \pi}{192} \approx 5000$ p kg/m^3

$\omega^2 R = \frac{2g}{3} = \frac{2}{3}\frac{GM}{R^2}$

$\omega^2 (R + h) = \frac{GM}{(R+h)^2}$

$\left(1 + \frac{h}{R}\right) = \frac{3}{2}\frac{R^2}{(R+h)^2} \Rightarrow 1 + h = \sqrt{\frac{3}{2}}$

$\Rightarrow h = \left(\sqrt{\frac{3}{2}} - 1\right) R$

$gh = g - \omega^2 R \cos 2\theta$

$= g - \frac{2g}{3} \times \frac{1}{4} = \frac{5g}{6}$

4) $\lambda = \frac{1240}{50 \times 10^3}$ nm

$f = \frac{c}{\lambda} = \frac{3 \times 10^8 \times 50 \times 10^3}{1240 \times 10^{-9}} = \frac{150}{124} \times 10^{49} = 1.2 \times 10^{19}$Hz

minimum wavelength

$\lambda_{min} = \frac{1240}{50 \times 10^{+3}} = 24.8 \times 10^{-3}$nm

Rate of emission of electron

$\frac{dN}{dt} = \frac{20 \times 10^{-3}}{1.6 \times 10^{-19}} = 1.25 \times 10^{17}$/sec

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6) $\lambda = \frac{h}{mv}$

$\frac{dh}{dt} = -\frac{h}{mv^2}\frac{dV}{dt}$

$10^{-4} = -\frac{10^{-34} \times 6.6}{6.6 \times 10^{-30}}\frac{1}{V^2}\frac{dV}{dt}$

$\frac{dV}{dt} = -V^2$

$\int_{10}^{V} \frac{dV}{V^2} = -\int_{0}^{t} dt$

$\frac{1}{V} - \frac{1}{10} = t$

$\frac{1}{V} = \frac{1+10t}{10}$

$V = \frac{10}{1+10t}$

at t = 0.9 s V = 1 m/s

Also $a = -V^2$

$|a| = 1\ m/s^2$

7) For (I): Since no heat transfer takes place $\Delta Q = 0$. As temperature of block increases $\Delta U > 0$, and by first law of thermodynamics $\Delta W < 0$

For (II): Again $\Delta Q = 0$, but $\Delta U > 0$ then $\Delta W < 0$

For (III): Isobaric expansion : $\Delta Q > 0$, $\Delta W > 0$, $\Delta U > 0$

For (IV): PT^3 = constant, P^4V^3 = constant, $PV^{3/4}$ = constant

$C = \frac{\left(\frac{5}{3}-\frac{3}{4}\right)R}{\left(\frac{5}{3}-1\right)\left(1-\frac{3}{4}\right)} > 0$ as P increase ↓ so $\Delta Q < 0$

$dW = \frac{RdT}{1-n} = \frac{RdT}{1-\frac{3}{4}}$

as T ↓ so $\Delta W < 0$ and T ↓ so ΔU ↓ 0

8) $_{91}Pa^{230} \rightarrow {}_{90}Th^{230} + {}_{1}e^{0} + \nu$ (β^+ decay)

$_{30}Zn^{57} \rightarrow {}_{29}Cu^{56} + {}_{1}P^{1}$ (proton decay)

$_{1}H^{3} \rightarrow {}_{2}He^{3} + {}_{-1}e^{0} + \bar{\nu}$ ($\beta^{\ominus}$ decay)

$_{94}Pu^{239} \rightarrow {}_{92}U^{235} + {}_{2}He^{4}$ (α-decay)

9) (A)

Near Q_1, potential is positive ⇒ Q_1 is +ve

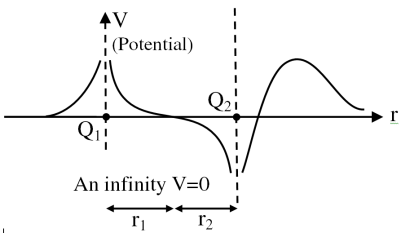
Near Q_2, potential is positive ⇒ Q_2 is –ve

For V = 0 & $r_1 = r_2$

$\frac{kQ_1}{r_1} - \frac{kQ_2}{r_2} = 0$

$\frac{|Q_1|}{r_1} = \frac{|Q_2|}{r_2}$

$\frac{|Q_1|}{|Q_2|} = \frac{r_1}{r_2} > 1$



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$|Q_1| > |Q_2|$

(B)

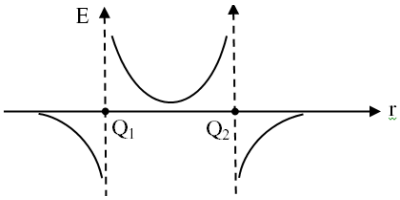
E_{min} is at mid point $\Rightarrow |Q_1| = |Q_2|$

Near Q_1 : $\xleftarrow{-E}$ Q_1 $\xrightarrow{+E}$

So, Q_1 is +ve

Near Q_2 : $\xrightarrow{+E}$ Q_2 $\xleftarrow{-E}$

So, Q_2 is –ve



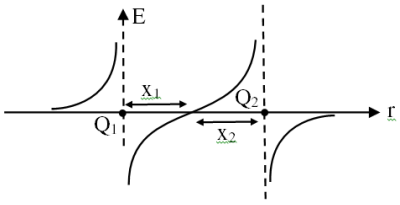
(C)

Near Q_1 : $\xrightarrow{+E}$ Q_1 $\xleftarrow{-E}$

So, Q_1 is –ve

Near Q_2 : $\xrightarrow{+E}$ • $\xleftarrow{-E}$

So, Q_2 is –ve



for $E = 0$; $x_1 < x_2$

$$\frac{kQ_1}{x_1^2} - \frac{kQ_2}{x_2^2} = 0$$

$$\frac{|Q_1|}{|Q_2|} = \frac{x_1^2}{x_2^2} < 1$$

$$|Q_1| < |Q_2|$$

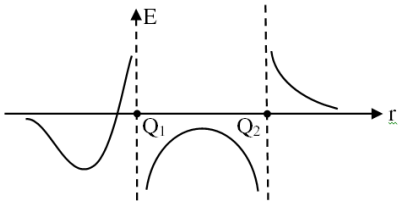
(D)

Near Q_1 : $\xrightarrow{+E}$ Q_1 $\xleftarrow{-E}$

So, Q_1 is –ve

Near Q_2 : $\xleftarrow{-E}$ Q_2 $\xrightarrow{+E}$

So, Q_2 is +ve



$E = 0$ is near Q_1

$E_1 - E_2 = 0$

$|E_1| = |E_2|$

$$\frac{kQ_1}{r_1^2} = \frac{kQ_2}{r_2^2}$$

$$\frac{|Q_1|}{|Q_2|} = \left(\frac{r_1}{r_2}\right)^2 < 1$$

$$|Q_1| < |Q_2|$$

10) (A) $\frac{\Delta Q}{\Delta t} = \frac{k_0[\pi(2R)^2 - \pi R^2]}{3R}(T_1 - T_2)$

$= \pi R k_0 (T_1 - T_2)$

(B) $\frac{\Delta Q}{\Delta t} = \frac{4\pi k R_1 R_2}{(R_2 - R_1)}(T_1 - T_2)$

$= x = \frac{4\pi k_0 3R \cdot R}{(3R - R)}(T_1 - T_2)$

$= 6\pi k_0 R(T_1 - T_2)$

(C) $\frac{\Delta Q}{\Delta t} = \frac{\frac{2\pi k_0 l}{\ln\left(\frac{R_2}{R_1}\right)}(T_1 - T_2)}{} = \frac{4\pi k_0 R}{\ln 2}(T_1 - T_2)$

(D) $\frac{\Delta Q}{\Delta t} = -\pi R^2 k \cdot \frac{dT}{dx} = -\pi R^2 k^0 \left(1 + \frac{x}{3R}\right)\frac{dT}{dx}$

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$$\therefore \frac{\Delta Q}{\Delta t} = \int_0^{3R} \frac{dx}{\left(1+\frac{x}{3R}\right)} = -\pi R^2 k \int_{T_1}^{T_2} dt$$

$$\frac{\Delta Q}{\Delta t} = \frac{\ln\left(1+\frac{x}{3R}\right)}{(R/3)}\Bigg|_0^{3R} = \pi R^2 k\,(T_1 - T_2)$$

$$\therefore \frac{\Delta Q}{\Delta t} = \frac{\pi k_0 R(T_1 - T_2)}{3\ln 2}$$

11) $\lambda_{th} = \frac{1240}{2} = 620$

$$i = \frac{dN}{dt}e = \frac{e \times A}{hc}\int \lambda dl = \frac{eAk\int_{400}^{620} \lambda d\lambda}{hc} = \frac{eAk\lambda^2}{2\lambda c}\Bigg|_{400}^{620}$$

$$\frac{1 \times 1 \times \left(620^2 - 400^2\right) \times k}{1240 \times 2}$$

$$= \frac{220 \times 1020}{1240 \times 2} \times \frac{0.031}{100} = 2.8A \approx 3A$$

12) Using the energies for the photons for two given transitions,

$$E_{2n} - E_n = 40.8 \Rightarrow 13.6z^2\left[\frac{1}{n^2} - \frac{1}{4n^2}\right] = 40.8$$

and

$$E_{2n} - E_1 = 204 \Rightarrow 13.6z^2\left[\frac{1}{1^2} - \frac{1}{4n^2}\right] = 204$$

13) Let N_0 be the initial number of nuclei, then

$N_1 = N_0 e^{-\lambda t_1}$

and $N_2 = N_0 e^{-\lambda t_2}$

$\therefore$ number of nuclei decayed $= N_1 - N_2$

$= N_0(e^{-\lambda t_1} - e^{-\lambda t_2})$

$$= \frac{A_0}{\lambda}(e^{-\lambda t_1} - e^{-\lambda t_2})$$

$$= \frac{A_1 - A_2}{\lambda} = (A_1 - A_2)\tau.$$

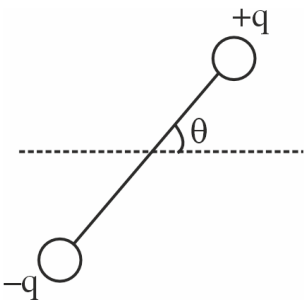
14) $0.6 = \frac{\text{work done}}{Q_{input}} = \frac{Q_{input} - Q_{reject}}{Q_{input}} = 1 - \frac{Q_r}{Q_i}$

$$0.6 = 1 - \frac{20}{Q_i}$$

$Q_i = 50$

$W = Q_i - Q_r = 30$ J

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15) $\frac{2kq}{R} = E\ell\cos\theta$.

$q = \frac{RE\ell\cos\theta}{2k}$

P (dipole) = $\frac{RE\ell^2\cos\theta}{2k}$

$\vec{\tau} = \vec{P} \times \vec{E}$

$\tau = I\alpha$

$\alpha = -w^2\theta$

Find ω hare and time period

16) Applying ν = u + at in y direction

$100\cos 30° \tan 60° = 100 \sin 30° + \frac{eE}{m}t$

{ at entry point and exit point $\nu_x = 100\cos 30°$

At exit = $\frac{\nu_y}{100\cos 30°} = \tan 60°$

$\Rightarrow \nu_y = 100\cos 30° + \tan 60°$}

$\frac{eE}{m}t = 100 \times \frac{\sqrt{3}}{2}.\sqrt{3} - 100 \times \frac{1}{2} = 100$

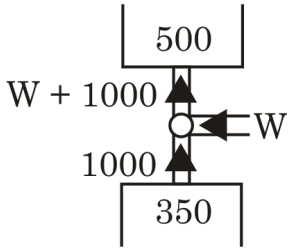
$\Rightarrow \frac{e}{m}E.\frac{m}{e} = 100$

17) $\frac{dH}{dt} = -2\pi x^2 k\frac{dT}{dx}$

$\Rightarrow \frac{dH}{dt} \times \int_{2.5}^{3} \frac{dx}{x^2} = 2\pi k \int_{-40}^{5} dT$

$\frac{dH}{dt} = \frac{2\pi \times k \times 45}{\frac{1}{2.5} - \frac{1}{3}} = 2.25\pi kW$

$2.25 \times 3.14 = 7.065$ kW



18) $COP = \frac{T_H}{T_H - T_L} = \frac{W + 1000}{W}$

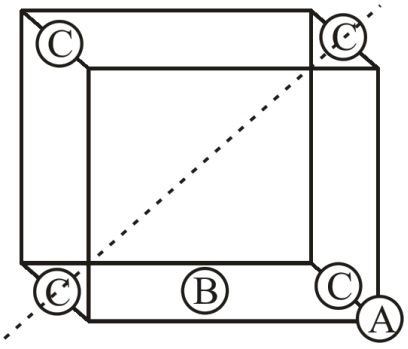
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$$\frac{500}{150} = 1 + \frac{1000}{W}$$

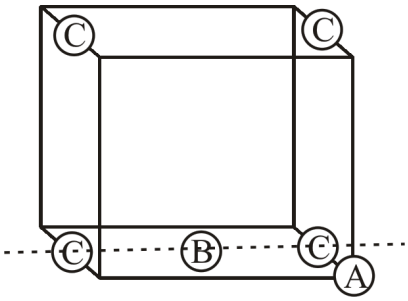
W = 428.57 J

PART-2 : CHEMISTRY

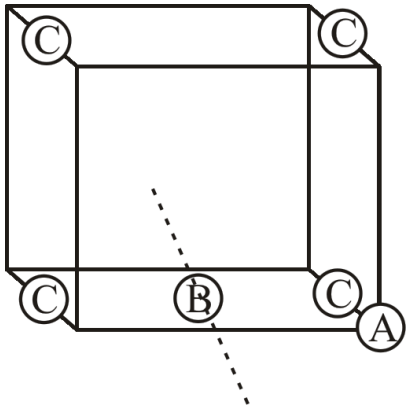
19)

(B) 

$A_{8\times1/8}\,B_{1/2\times2}\,C_{2\times1/4} \equiv A_2B_2C$

(C) 

$A_{8\times1/8}\,B_{1/2\times1}\,C_{2\times1/4} \equiv A_2BC$

(D) 

$A_{8\times1/8}\,B_{1/2\times1}\,C_{4\times1/4} \equiv A_2BC_2$

20) (A) Number of nodal planes = □ = 2 and radial nodes = n – □ – 1 = 5 – 2 – 1 = 2

(B) Spin quantum number does not obtain from schrodinger wave equation when applied to an atom.

(C) 5 → 1 transition for 1 electron, maximum number of photons that may be emitted are 4.
5 → 4, 4 → 3, 3 → 2, 2 → 1

(D) If Shortest wavelength of H-atom in Lyman series ∞ → 1, $\lambda - x\frac{912}{(1)^2\left(\frac{1}{1}-\frac{1}{\infty}\right)}A$

Longest wavelength in Balmer series of He^+ ion 3 → 2, $\lambda = \frac{912}{(2)^2\left(\frac{1}{4}-\frac{1}{9}\right)}A$

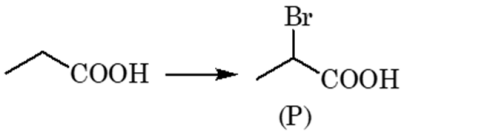
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21) Bond strength of CO < CO^+.
Bond length ∝ 1/bond order.
Unpaired electrons : O_2 = 2, NO = 1 and CO = 0
Number of ABMO electrons : O_2 = 6, N_2^- = 5 and Be_2 = 4.

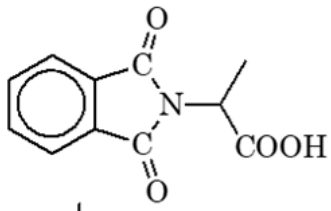
22) Cl_2 is a good oxidising agent. It can oxidise
Fe^{2+} to Fe^{3+}; SO_3^{2-} to SO_4^{2-}; SO_2 to SO_4^{2-} and I_2 to IO_3^- etc.

23)

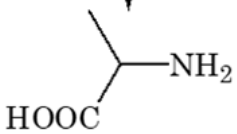
(A) is wrong



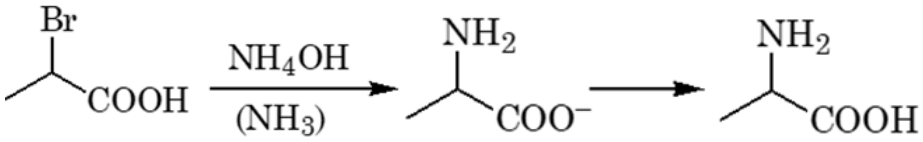
(P)



Hydrolysis
(i)OH^-, (ii)H_3O^+



(Q)

(B)


(C) Due to presence of $-NH_2$ group $NaNO_2$ + HCl liberate N_2 gas
(D) Due to –I effect of Br.

24)

All correct.

26) Pt, H_2(g, 1 atm) | H^+(aq. 0.1M) || Cl^-(aq. 0.01 M) | Cl_2(g, 1 atm)
Cell reaction : $H_2(g) + Cl_2(g) \rightarrow 2H^+(aq) + 2Cl^-(aq)$

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$$E = E^\circ - \frac{0.06}{2}\log\frac{[H^+]\,[Cl^-]^2}{P_{H_2} \times P_{Cl_2}}$$

(P) $\Delta E = E_2 - E_1 = -\dfrac{0.06}{2}\log\dfrac{(10^{-5})^2}{(0.1)^2} = 0.24$ V

(Q) $\Delta E = E_2 - E_1 = -\dfrac{0.06}{2}\log\dfrac{(0.1)^2}{(0.01)^2} = -0.06$ V

(R) $\Delta E = E_2 - E_1 = -\dfrac{0.06}{2}\log\dfrac{1}{10} = 0.03$ V

(S) $\Delta E = E_2 - E_1 = -\dfrac{0.06}{2}\log\dfrac{1}{0.1} = -0.03$ V

27) (P) $P_2O_3 + 3H_2O \rightarrow 2\underline{H_3PO_3}$

(Q) $P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + \underline{PH_3}$

(R) $PCl_5 + CH_3COOH \rightarrow CH_3COCl + \underline{POCl_3} + HCl$

(S) $H_3PO_2 + 2H_2O + 4AgNO_3 \rightarrow 4Ag + 4HNO_3 + \underline{H_3PO_4}$

29) $2KI + HgI_2 \rightarrow K_2[HgI_4]$

0.2m 0.1m 0.1m

$$\frac{(\Delta T_f)_1}{(\Delta T_f)_2} = \frac{i_1 m_1}{i_2 m_2}$$

$$\frac{0.2}{(\Delta T_f)_2} = \frac{2 \times 0.2}{3 \times 0.1}$$

$(\Delta T_f)_2 = 0.15$

T_f (solution) 273 – 0.15 = 272.85 K

30) Suppose 1 mole of A produced x mole of AB(g) and (1 – x) mole of AB_4

140 = 100x + 200(1 – x)

x = 0.6

Moles of B_2(g) used = $\dfrac{1}{2} \times 0.6 + 2 \times (1 - 0.6) = 1.10$

32) XeF_2, XeF_4

35)

$COO^{\ominus}$ and NH_2 are the two basic group.

36)

i, iii, iv, vi

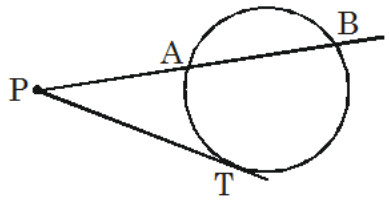
PART-3 : MATHEMATICS

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37) tangent at (x_1y_1) $\frac{xx_1}{4} - \frac{yy_1}{2} = 1$

compare with $x + \frac{\sqrt{6}y}{2} = 1$

$x_1 = 4,\ y_1 = -\sqrt{6}$



38)

$PA.PB = (PT)^2$

So, $\frac{PA + PB}{2} \geq \sqrt{PA.PB} = PT = \sqrt{S_1}$

$\Rightarrow PA + PB \geq 6$

39) (A) $O \times Q^c = 0$

(B) $\sqrt{2} + \left(-\sqrt{2}\right) = 0$

(C) $a^2 + b^2 + c^2 - ab - bc - ca$

$= \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2]$

(D) $(a + b + c) = 0$

$\Rightarrow (ab + bc + ca)^2 = (ab)^2 + (bc)^2 + (ca)^2$

40) $|adj(kA)| = |kA|^2 = (k^3|A|)^2 = k^6|A|^2$

$|adj(adj\ (kA))| = |kA|^4 = (k^3|A|)^4 = k^{12}|A|^4$

$\left|\frac{A}{|A|}\right| = \frac{1}{|A|^3}|A| = \frac{1}{|A|^2}$

$\left|(kA)^{-1}\right| = \frac{1}{|kA|} = \frac{1}{k^3|A|}$

41) $\frac{4x}{\cos\theta} - \frac{2y}{\sin\theta} = 12$

A(3cosθ, 0) B(0, -6sinθ)

$2h = 3\cos\theta,\quad 2k = -6\sin\theta$

$\Rightarrow \frac{4x^2}{9} + \frac{4y^2}{36} = 1$

$4x^2 + y^2 = 9$

(C) $\frac{x^2}{16} + \frac{y^2}{4} = 1$, $e = \sqrt{1 - \frac{4}{16}} = \frac{\sqrt{3}}{2}$

$F_1\left(2\sqrt{3},\ 0\right)$ & $F_2\left(-2\sqrt{3},\ 0\right)$

$B_1(0, 2)$ & $B_2(0, -2)$

$A = \frac{1}{2} \times 4\sqrt{3} \times 4 = 8\sqrt{3}$

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(D) $\frac{x\cos\theta}{4}+\frac{y\sin\theta}{2}=1$

L = (4secθ, 0) M = (0, 2cosecθ)

$LM=\sqrt{16\sec^2\theta+4\operatorname{cosec}^2\theta}$

$=\sqrt{20+16\tan^2\theta+4\cot^2\theta}$

$LM_{min}=6$

42) $2^{2\sin^2x-3\sin x+1}+2^{3\sin x-2\sin^2x+2}=9$

$\Rightarrow 2^{2\sin^2x-3\sin x+1}+2^{2-(2\sin^2x-3\sin x)}=9$

$2^{2\sin^2x-3\sin x}=t$

$2t+\frac{4}{t}=9 \Rightarrow t=4 \text{ or } t=\frac{1}{2}$

$\Rightarrow 2^{2\sin^2x-3\sin x}=2^2$ or $2^{2\sin^2x-3\sin x}=2^{-1} \Rightarrow 2\sin^2x-3\sin x=2$

or $2\sin^2x-3\sin x=-1$

$\Rightarrow \sin x=-\frac{1}{2},\ 1,\ \frac{1}{2},\ 2$ (reject)

5 solutions in [0, 2π]

43) (I) $\begin{vmatrix}2&-1&1\\1&-2&1\\\lambda&-1&2\end{vmatrix}=0$

$\Rightarrow 2(-3)+(2-\lambda)+1+2\lambda=0$

$-6+2-\lambda-1+2\lambda=0$

$\lambda=5$

(II) $\begin{vmatrix}1&1&1\\(C+2)&(C+4)&6\\(C+2)^2&(C+4)^4&36\end{vmatrix}=0$

$C_2-C_2-C_1$

$\begin{vmatrix}1&0&1\\(C+2)&2&6\\(C+2)^2&(12+4C)&36\end{vmatrix}=0$

$\Rightarrow C=2,4$

(III) Use $abc+2fgh-af^2-bg^2-ch^2=0$

$p=2 \quad \& \quad q=3$

$p=2 \quad \& \quad q=\frac{-9}{2}$

(IV) $x^2+y^2-\frac{5}{2}x+3y-\frac{1}{2}=0$

Centre is $\left(\frac{5}{4},-\frac{3}{2}\right)$

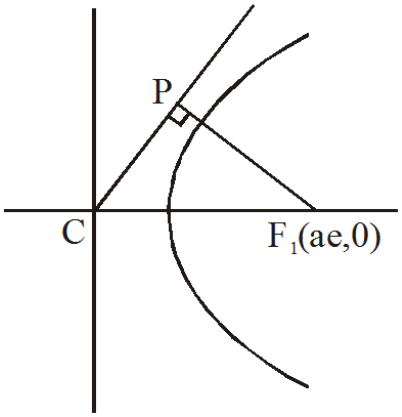
$\frac{5\lambda}{4}-\frac{3\mu}{2}=1$

$5\lambda-6\mu=4$

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44) Length of perpendicular from $F_1(ae, 0)$ upon $bx - ay = 0$ is equal to

$$F_1P = \left|\frac{b(ae)-0}{\sqrt{a^2+b^2}}\right| = \frac{b\sqrt{a^2+b^2}}{\sqrt{a^2+b^2}} = b$$



$\Rightarrow$ Triangle CF_1P is right angle isosceles

$\Rightarrow$ $CP = F_1P = b$ and $CF_1^2 = CP^2 + PF_1^2$

$\Rightarrow$ $a^2e^2 = 2b^2 \Rightarrow a^2e^2 = 2(a^2e^2 - a^2)$

$\Rightarrow$ $e = \sqrt{2} \Rightarrow a = b$

45) (I) $(x + y + \lambda_1)(x + y + \lambda_2) = 0$

$x^2 + y^2 + 2xy + x(\lambda_1 + \lambda_2) + y(\lambda_1 + \lambda_2) + \lambda_1\lambda_2 = 0$

$\lambda_1 + \lambda_2 = -8\lambda$, $\lambda_1\lambda_2 = -9\lambda^2$

$$25\sqrt{2} = \left|\frac{\lambda_1 - \lambda_2}{\sqrt{2}}\right| = 50 = \sqrt{64\lambda^2 - 4(-9\lambda^2)} = 10|\lambda|$$

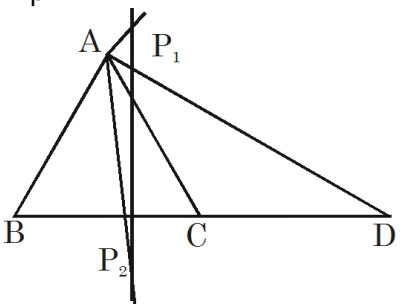
$\Rightarrow |\lambda| = 5$

(II) $\lambda(\lambda - 3) + \lambda + 1 = 0$

$\lambda^2 - 2\lambda + 1 = 0 \Rightarrow \lambda = 1$

(III) $A(1,2) \rightarrow B(2,1)$ ® $C(2,-1)$ ® $D(1,-2)$

$\alpha + \beta = 1$



(IV)

P set of points lies on perpendicular bisector of BC

Q set of point lies on angle bisector

$n(P \cap Q) = 2$

46) $y^2 - 4y - x + 3 = 0$

$\Rightarrow (y - 2)^2 - 4 - x + 3 = 0$

$(y - 2)^2 = x + 1$

$Y^2 = X$

$Y = y - 2$ & $X = x + 1$

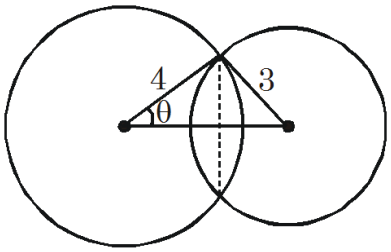
$a = \frac{1}{4}$

	X,Y	(x,y)
Vertex	(0,0)	(–1,2)
L.R.	1	1

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focus	$\left(\frac{1}{4},0\right)$	$\left(-\frac{3}{4},2\right)$
foot of directrix	$\left(-\frac{1}{4},0\right)$	$\left(-\frac{5}{4},2\right)$
extemity or L.R.	$\left(\frac{1}{4},\frac{1}{2}\right)$	$\left(-\frac{3}{4},\frac{5}{2}\right)$
	$\left(\frac{1}{4},-\frac{1}{2}\right)$	$\left(-\frac{3}{4},\frac{3}{2}\right)$

47) $\tan\theta = \frac{3}{4}$



□ = 2(4sinθ)

$= 8\frac{3}{5} = \frac{24}{5}$

$\frac{24}{\ell} = 5$

48) $\text{adjA} = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$, $|A| > 0$

$|adjA| = |A|^2 = 16$

or $|A| = 4$

□ adj(adjA) = |A|.A = 4A

then diagonal elements of A

49) 2785x

Sum of digit at ODD places = 10 + x

Sum of digit at even places = 12

Divisible by 11

□ (2 – x) = 0 □ x = 2

50) □ = $4\log_3 4.\log_2 9$

$\ell = \frac{4\log 4}{\log 3}.\frac{\log 9}{\log 2} = 16$

$m = \frac{4}{5}\left(1+9^{\log_3 8}\right)^{\log_{65}5}$

$m = \frac{4}{5}\left(1+3^{2\log_3 8}\right)^{\log_{65}5}$

$m = \frac{4}{5}(65)^{\log_{65}5} = 4$

$2^{(\log_2 3)^x} = 3^{(\log_3 2)^x}$

$\Rightarrow (\log_2 3)^x \log 2 = (\log_3 2)^x \log 3$

$(\log_2 3)^{2x} = (\log_2 3)$

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$2x = 1 \Rightarrow x = \frac{1}{2}$

51) $A^2 = \begin{bmatrix} 1 & 4 & 2a+8 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & 6 & 3a+24 \\ 0 & 1 & 12 \\ 0 & 0 & 1 \end{bmatrix}$

:

:

$\Rightarrow A^n = \begin{bmatrix} 1 & 18 & 2007 \\ 0 & 1 & 36 \\ 0 & 0 & 1 \end{bmatrix}$

$\Rightarrow 2n = 18 \Rightarrow n = 9$

$9a + 8\sum_{k=0}^{8} k = 2007$

$\Rightarrow 9a + 8\left[\frac{8 \times 9}{2}\right] = 2007$

$\Rightarrow 9a = 2007 - 288$

$\Rightarrow 9a = 1719 ; a = 191$

$\therefore n + a = 9 + 191 = 200$

52) $LHS = \begin{vmatrix} 1 & (\cos\alpha - \sin\alpha) & (co\alpha + \sin\alpha) \\ 1 & (\cos\beta - \sin\beta) & (\cos\beta + \sin\beta) \\ 1 & (\cos\gamma - \sin\gamma) & (\cos\gamma + \sin\gamma) \end{vmatrix}$

$C_2 \to C_2 + C_3$

$2\begin{vmatrix} 1 & \cos\alpha & (\cos\alpha + \sin\alpha) \\ 1 & \cos\beta & (\cos\beta + \sin\beta) \\ 1 & \cos\gamma & (\cos\gamma + \sin\alpha) \end{vmatrix}$

$C_3 \to C_3 - C_2$

53) $y = mx + c$ is tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

if $c^2 = a^2m^2 - b^2$

$\therefore (2\sqrt{5})^2 = 9m^2 - 16$

$36 = 9m^2$

$m^2 = 4 \quad \therefore \quad m = 2, -2$

sum of roots = a + b = 0

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product of roots = k = –4
a + b + k = –4

54) For least area, circle must touch the ellipse solving the two equations

$$\frac{x^2}{a^2} + \frac{1-(x-1)^2}{b^2} = 1$$

$(b^2 - a^2)x^2 + 2a^2x - a^2b^2 = 0$
D = 0
$\Rightarrow a^2 + b^2(b^2 - a^2) = 0$ .....(1)
Also $b^2 = a^2(1 - e^2)$ .....(2)
$\Rightarrow a^2 + b^2(-a^2e^2) = 0$ thus $b = \frac{1}{e}, a = \frac{1}{e\sqrt{1-e^2}}$
Area = $\pi ab = \frac{\pi.1}{e^2\sqrt{1-e^2}}$
$f(e) = e^4 - e^6$
$f'(e) = 4e^3 - 6e^5 = 0$
$2e^3(2 - 3e^2) = 0$
$e^2 = \frac{2}{3}$

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